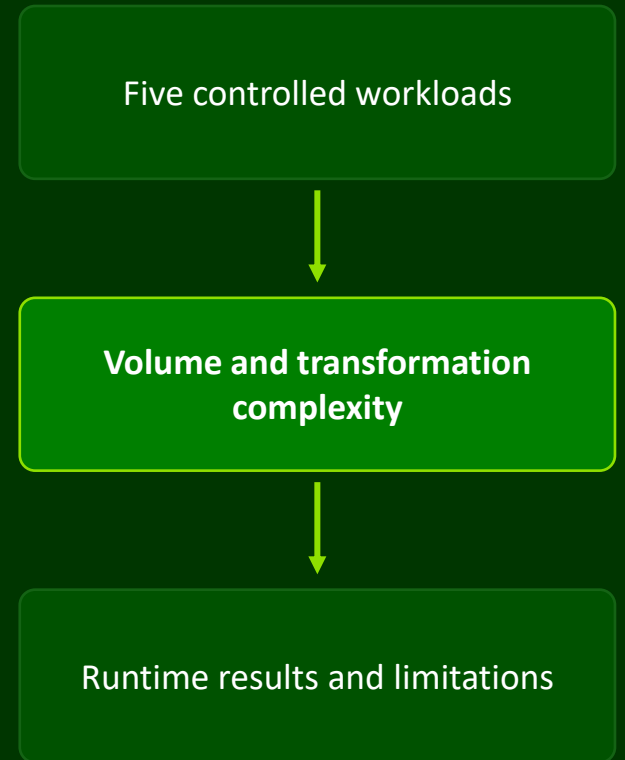


UPSTREAM LONG-TERM PLANNING

Performance and Scalability Testing Scenario Modelling Engine

Runtime evidence across five controlled workloads of increasing volume and transformation complexity



The testing approach

■ Performance and Scalability Testing

EXECUTIVE SUMMARY — SCENARIO MODELLING ENGINE

The engine processed the full representative planning workload as transformation complexity increased



31.33s

Mean runtime

Most transformation-intensive test



720

Case operations

Full Plan Book-aligned population, five scenarios



152,942

Output rows

Generated by the highest-transformation workload

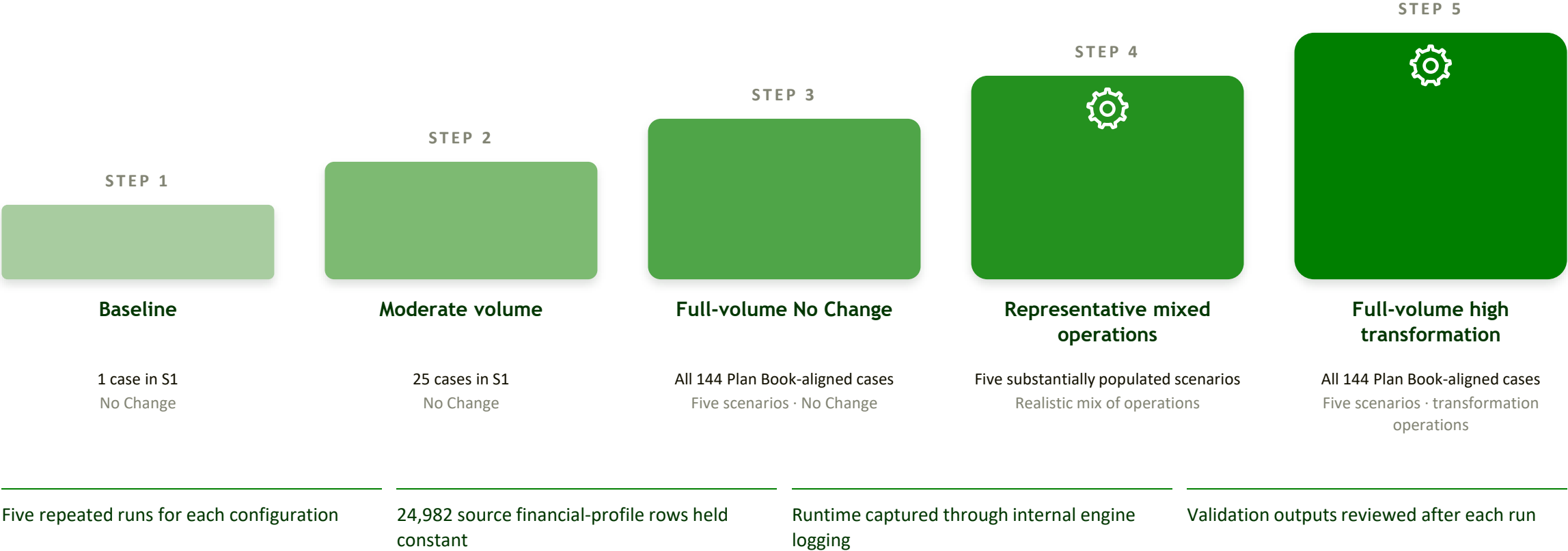
Performance testing covered the full 144-case modelling population aligned to the Plan Book activity cut.

The engine successfully processed this population across all five configured scenarios, demonstrating practical performance at a representative full-volume planning workload.

The testing evaluates the current Python-in-Excel prototype and does not assess enterprise infrastructure characteristics.

Five controlled workloads tested both volume and transformation complexity

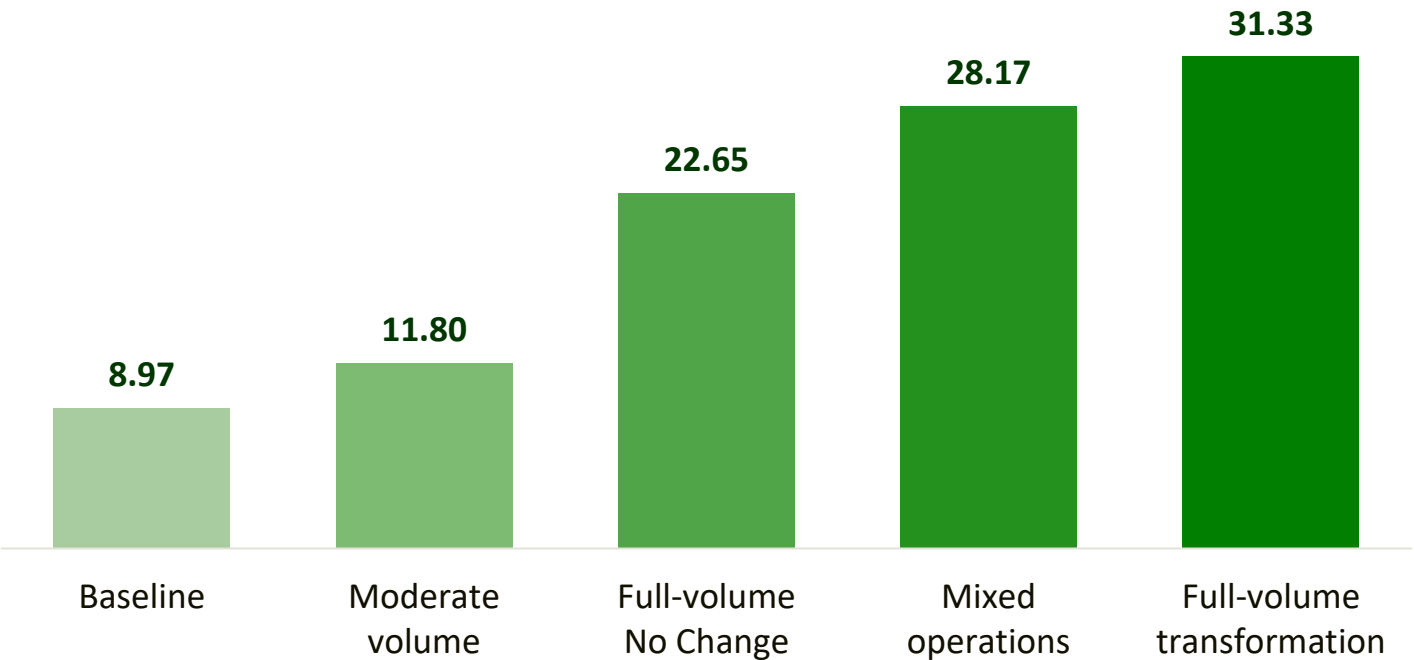
TEST DESIGN



The maximum-volume tests used all 144 available modelling cases across five scenarios, equivalent to 720 case operations. This represents the full Plan Book-aligned population used by the prototype rather than a test sample.

Runtime increased predictably and remained below 32 seconds on average

PERFORMANCE RESULTS — MEAN RUNTIME BY TEST



Mean of five runs per configuration, at full Plan Book-aligned volume for tests 3 and 5. Output-row counts are not directly comparable between tests, because transformation operations change the resulting row population.



+38%

Transformation-intensive processing increased mean runtime against the full-volume No Change test, remaining within a practical range for interactive modelling.



Full representative volume

The transformation-intensive workload processed 720 case operations and generated 152,942 output rows in an average of 31.33 seconds.

OBSERVED RUN RANGES (SECONDS)

Baseline	5.25–11.62
Moderate volume	11.51–12.14
Full-volume No Change	20.86–24.83
Mixed operations	25.12–33.10
Full-volume transformation	25.67–36.25

The prototype supports its full representative planning workload

CONCLUSION AND LIMITATIONS



Conclusion

- The prototype can support its full representative planning workload.
- Runtime remained practical with all Plan Book-aligned cases processed across five scenarios.
- Both No Change and transformation-intensive configurations completed successfully.
- The full-volume transformation workload averaged approximately 31 seconds.
- Runtime increased with volume and transformation complexity as expected.



Limitations

- Testing was performed within the current Python-in-Excel implementation.
- Runtime may be influenced by machine, session and Excel/Python environment conditions.
- Results demonstrate performance for the intended Plan Book-aligned planning workload.
- Greater depth of granularity (closer to financial entity granularity) would increase case number beyond tested limit.
- The tests do not assess enterprise infrastructure characteristics such as concurrent users, service availability or production deployment resilience.

The testing demonstrates fitness for the prototype's representative planning workload rather than enterprise infrastructure performance.

Detailed method, individual run results and supporting evidence are retained in the Performance and Scalability Testing document.

■ Testing evidence – screenshots of Excel application home page

Baseline

Model Engine Log

Run duration: 8.00 seconds

Scenario Configuration

Scenario (S1)

All data for testing

✖ Some cases not configured

1 (of 144) Cases Included | 1 Operations Configured | Distinct operations used: No Change [1]

Scenario (S2)

Select some cases to make a scenario

Empty scenario

Scenario (S3)

Select some cases to make a scenario

Empty scenario

Scenario (S4)

Select some cases to make a scenario

Empty scenario

Scenario (S5)

Select some cases to make a scenario

Empty scenario

Moderate Volume

Model Engine Log

Run duration: 11.95 seconds

Scenario Configuration

Scenario (S1)

All data for testing

✖ Some cases not configured

25 (of 144) Cases Included | 25 Operations Configured | Distinct operations used: No Change [25]

Scenario (S2)

Select some cases to make a scenario

Empty scenario

Scenario (S3)

Select some cases to make a scenario

Empty scenario

Scenario (S4)

Select some cases to make a scenario

Empty scenario

Scenario (S5)

Select some cases to make a scenario

Empty scenario

Full-volume No Change

Model Engine Log

Run duration: 20.86 seconds

Scenario Configuration

Scenario (S1)

All data for testing

✔ Scenario ready to run

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: No Change [144]

Scenario (S2)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: No Change [144]

✔ Scenario ready to run

Scenario (S3)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: No Change [144]

✔ Scenario ready to run

Scenario (S4)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: No Change [144]

✔ Scenario ready to run

Scenario (S5)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: No Change [144]

✔ Scenario ready to run

Mixed Operations

Model Engine Log

Run duration: 26.28 seconds

Scenario Configuration

Scenario (S1)

All data for testing

✔ Scenario ready to run

139 (of 144) Cases Included | 139 Operations Configured | Distinct operations used: No Change [131], Delay [3], Farm In (Increase) [2], Truncate

Scenario (S2)

133 (of 144) Cases Included | 133 Operations Configured | Distinct operations used: No Change [127], Delay [2], Farm In (Increase) [2], Truncate

✔ Scenario ready to run

Scenario (S3)

128 (of 144) Cases Included | 128 Operations Configured | Distinct operations used: No Change [122], Delay [2], Farm In (Increase) [2], Truncate

✔ Scenario ready to run

Scenario (S4)

143 (of 144) Cases Included | 143 Operations Configured | Distinct operations used: Delay [11], No Change [128], Farm In (Increase) [2], Truncate

✔ Scenario ready to run

Scenario (S5)

143 (of 144) Cases Included | 143 Operations Configured | Distinct operations used: No Change [137], Delay [2], Farm In (Increase) [2], Truncate

✔ Scenario ready to run

Full-volume High Transformation

Model Engine Log

Run duration: 32.58 seconds

Scenario Configuration

Scenario (S1)

All data for testing

✔ Scenario ready to run

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: No Change [144]

Scenario (S2)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: Delay [144]

✔ Scenario ready to run

Scenario (S3)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: Accelerate [144]

✔ Scenario ready to run

Scenario (S4)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: Farm In (Increase) [144]

✔ Scenario ready to run

Scenario (S5)

144 (of 144) Cases Included | 144 Operations Configured | Distinct operations used: Farm Down (Dilution) [144]

✔ Scenario ready to run